Divergent (Concave) lens questions.

Diverging lenses are used in doors to allow house occupants see who is outside. A man is standing 1 m away from such a lens. The lens has a focal length of 20 cm, How far from the lens will his image appear?

A diverging lens has a focal length of 20 cm. An object is placed 15 cm from the lens,

(i) calculate how far the image will be from the lens.

(ii) Is the image real or virtual.

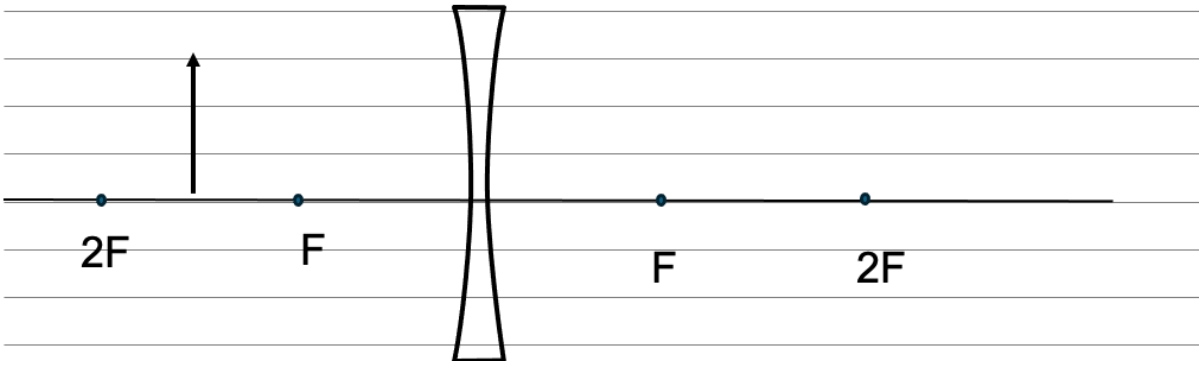


A diverging lens has a focal length of 20 cm.

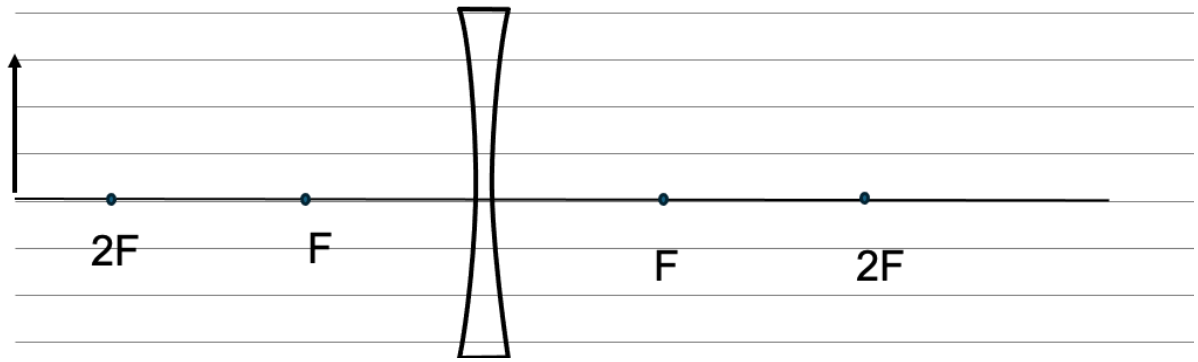
(i) What distance from the lens will an image of the moon form.

(ii) Justify your answer

Complete the Ray Diagram



Complete the Ray Diagram.

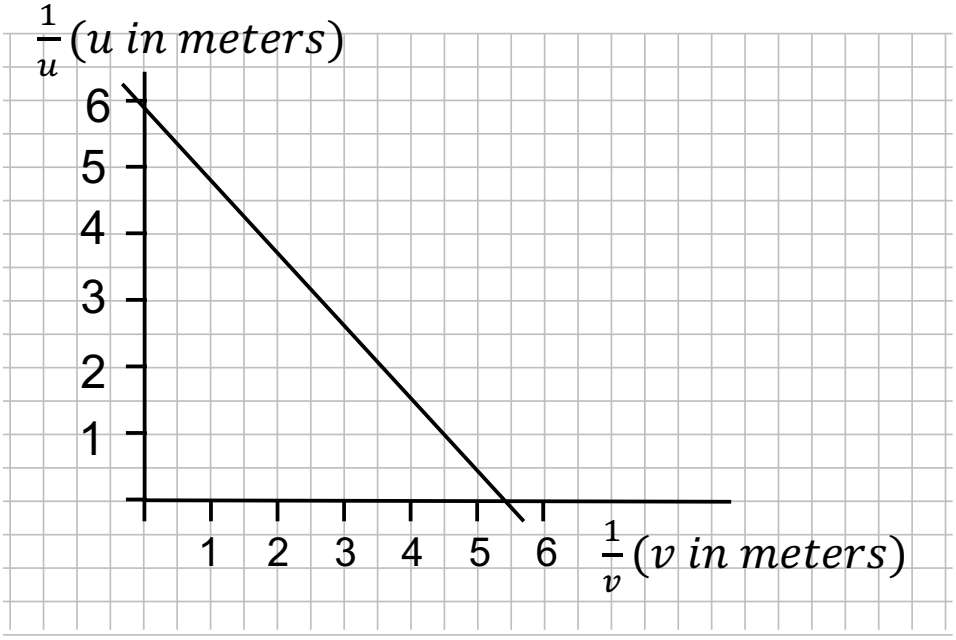


In an experiment to verify the lens formula $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$ for a diverging lens, u and v were measured for different arrangements (v was determined using the "no parallax" method). Use the data obtained to calculate the focal length of the lens at each data point. Use the calculations to check if the formula has been verified.

u (cm)	10.0	20.0	30.0	40.0	50.0
v (cm)	-6.1	-8.5	-10.0	-11.0	-11.4

Exercise with secondary data

In an experiment to verify the lens formula students obtained the following graph. Using the graph determine the focal length of the lens.



Exercise with secondary data

A student made measurements of u and v for the image of an object in a converging lens with f = 15 cm to verify the model $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$. Calculate the percentage error of each data point.

u (cm)	20	30	40
v (cm)	56.2	30.3	24.1